Python-এর **built-in functions** মানে হলো—যেগুলো ব্যবহার করতে আলাদা করে কোনো লাইব্রেরি import করতে হয় না। এগুলো সরাসরি Python-এ পাওয়া যায়।

🔹 1. Input / Output

```
print("Hello")
x = input("Enter number: ")
```

- `print()` → output দেখায়
- `input()` → user input নেয় (string হিসেবে)

🔹 2. Type Conversion

```
int("10")
float("3.5")
str(25)
bool(0)
list("abc")
tuple([1,2])
set([1,2,2])
```

- `int()`, `float()`, `str()`
- `bool()`
- `list()`, `tuple()`, `set()`

🔹 3. Numeric Functions

```
abs(-5)         # 5
pow(2, 3)       # 8
round(3.6)      # 4
min(1,2,3)
max(4,6,1)
sum([1,2,3])
```

- `abs()`
- `pow()`

- `round()`
- `min()`, `max()`
- `sum()`

🔹 4. Sequence & Collection

```
len([1,2,3])
sorted([3,1,2])
reversed([1,2,3])
enumerate(["a","b"])
zip([1,2],[3,4])
```

- `len()`
- `sorted()`
- `reversed()`
- `enumerate()`
- `zip()`

🔹 5. Type Checking

```
type(10)
isinstance(10, int)
```

- `type()`
- `isinstance()`

🔹 6. Logical / Boolean

```
all([True, True])
any([False, True])
```

- `all()` → সব True হলে True
- `any()` → একটি True হলেই True

🔹 7. Character / ASCII

```
ord('A')   # 65
chr(65)    # 'A'
```

- `ord()` → char → ASCII
- `chr()` → ASCII → char

🔹 8. Math / Iteration

```
range(5)
```

- `range()` → sequence তৈরি করে

🔹 9. Object / Memory

```
id(x)
help(print)
```

- `id()` → memory address
- `help()` → documentation

🔹 10. Execution / Dynamic

```
eval("2+3")
```

- `eval()` → string কে expression হিসেবে execute করে (⚠️ careful)

🔹 11. File / Advanced

```
open("file.txt")
```

- `open()` → file handle

🔹 Most Used Built-ins (Interview Favorite ⭐)

- `print()`
- `len()`
- `type()`
- `range()`
- `sum()`
- `min() / max()`
- `sorted()`
- `enumerate()`
- `zip()`
- `isinstance()`

Python-এর **built-in modules** মানে হলো—যেগুলো Python install করলেই পাওয়া যায়, আলাদা করে pip install করতে হয় না।

নিচে **সবচেয়ে important built-in modules** category অনুযায়ী দেওয়া হলো 👇

🔹 1. System & OS Related

`sys`

```
import sys
print(sys.argv)
print(sys.path)
sys.exit()
```

👉 command-line args, Python environment control

`os`

```
import os
os.getcwd()
os.listdir()
os.mkdir("test")
os.remove("file.txt")
```

👉 file, directory, environment variable

🔹 2. Math & Numbers

`math`

```
import math
math.sqrt(16)
math.ceil(4.2)
math.floor(4.9)
math.pi
```

`random`

```
import random
random.randint(1, 10)
random.choice([1,2,3])
random.shuffle(list)
```

🔹 3. Date & Time

`datetime`

```
from datetime import datetime
now = datetime.now()
print(now.year)
```

`time`

```
import time
time.sleep(2)
```

🔹 4. Data Structures

`collections`

```
from collections import Counter, defaultdict
Counter("banana")
```

`heapq`

```
import heapq
heapq.heappush(h, 5)
```

🔹 5. Iteration & Functional

`itertools`

```
import itertools
itertools.permutations([1,2,3])
```

`functools`

```
from functools import reduce
reduce(lambda x,y: x+y, [1,2,3])
```

🔹 6. File & Data Handling

`json`

```
import json
json.dumps({"a":1})
json.loads('{"a":1}')
```

`pickle`

```
import pickle
pickle.dump(obj, file)
```

`csv`

```
import csv
```

🔹 7. Text & Pattern

`re`

```
import re
re.search("ab", "abc")
```

`string`

```
import string
string.ascii_letters
```

🔹 8. Security & Encoding

`hashlib`

```
import hashlib
hashlib.md5(b"text").hexdigest()
```

`base64`

```
import base64
```

🔹 9. Debugging & Testing

`logging`

```
import logging
logging.warning("Warning!")
```

`unittest`

```
import unittest
```

🔹 10. Networking & Internet

`socket`

```
import socket
```

`http`

```
import http.client
```

🔹 11. Concurrency

`threading`

```
import threading
```

`multiprocessing`

```
import multiprocessing
```

🔹 12. Import Utilities

`importlib`

```
import importlib
```

⭐ Most Important Built-in Modules (Interview)

- `os`
- `sys`
- `math`
- `random`
- `datetime`
- `collections`
- `itertools`
- `json`
- `re`
- `logging`

🔹 Check All Built-in Modules

```
help("modules")
```

`datetime` module Python-এর **সবচেয়ে important built-in module**গুলোর একটি।
এখানে আমি **interview + real project-এ সবচেয়ে বেশি ব্যবহৃত method গুলো** সুন্দরভাবে explain

করছি 👇

📦 `datetime` Module (Important Methods)

```
import datetime
from datetime import date, time, datetime, timedelta
```

🔹 1️⃣ `date` Class (Only Date)

✔️ `date.today()`

আজকের তারিখ

```
from datetime import date
print(date.today())
```

✔️ `date(year, month, day)`

নিজে date বানানো

```
d = date(2026, 1, 20)
print(d)
```

✔️ `date.year`, `month`, `day`

```
print(d.year, d.month, d.day)
```

🔹 2️⃣ `time` Class (Only Time)

✔️ `time(hour, minute, second)`

```
from datetime import time
t = time(10, 30, 45)
print(t)
```

✔️ `hour`, `minute`, `second`

```
print(t.hour)
```

🔹 3️⃣ `datetime` Class (Date + Time) ⭐ MOST IMPORTANT

✔️ `datetime.now()`

বর্তমান date + time

```
from datetime import datetime
now = datetime.now()
print(now)
```

✔️ `datetime.today()`

`now()` এর মতোই কাজ করে

```
datetime.today()
```

✔️ `datetime(year, month, day, hour, min, sec)`

```
dt = datetime(2026,1,20,10,30,0)
```

✔️ Access Parts

```
now.year
now.month
now.day
now.hour
now.minute
now.second
```

🔹 4️⃣ Format & Parse (🔥 VERY IMPORTANT)

✔️ `strftime()` → date → string

```
now.strftime("%Y-%m-%d %H:%M:%S")
```

Code	Meaning
%Y	Year
%m	Month
%d	Day
%H	Hour (24)
%M	Minute
%S	Second
%A	Day name
%B	Month name

✔️ `strptime()` → string → datetime

```
dt = datetime.strptime("2026-01-20", "%Y-%m-%d")
```

📌 **Interview Question:**
➡️ *Difference between* `strftime` & `strptime`

- `strftime` → format
- `strptime` → parse

🔹 5️⃣ `timedelta` (Date Difference) ⭐⭐

✔️ Add / Subtract Date

```
from datetime import timedelta

future = now + timedelta(days=5)
past = now - timedelta(days=2)
```

✔️ Difference Between Two Dates

```
d1 = datetime(2026,1,1)
d2 = datetime(2026,1,20)
diff = d2 - d1
print(diff.days)
```

🔹 6️⃣ Comparison

```
if d1 < d2:
    print("Earlier")
```

🔹 7️⃣ Convert Between Types

✔️ datetime → date / time

```
now.date()
now.time()
```

🔹 8️⃣ ISO Format (API / Django)

```
now.isoformat()
```

🔹 9️⃣ Timestamp (Unix Time)

```
now.timestamp()
```

🔹 1🔟 Weekday / Calendar

```
now.weekday()       # 0 = Monday
now.isoweekday()    # 1 = Monday
```

⭐ Interview Most Important Summary

Topic	Why
`datetime.now()`	current time
`strftime()`	formatting
`strptime()`	parsing

Topic	Why
`timedelta`	date calculation
`timestamp()`	unix time
comparison	validation

🔥 Django / Real-World Usage

```
from django.utils import timezone
timezone.now()
```

⭐ Python `datetime` – Important Interview Questions

1️⃣ `datetime.now()` vs `datetime.today()`

Q: পার্থক্য কী?

A:
➡️ বাস্তবে কোনো পার্থক্য নেই
➡️ দুটোই current date + time দেয়

```
datetime.now()
datetime.today()
```

2️⃣ `date` vs `datetime`

Q: `date` আর `datetime` এর পার্থক্য?

A:

- `date` → শুধু date (year, month, day)
- `datetime` → date + time

3️⃣ `strftime()` vs `strptime()` ⭐⭐

Q: Difference?

A:

- `strftime()` → datetime → string (format)
- `strptime()` → string → datetime (parse)

4️⃣ `timedelta` কী কাজে লাগে?

A:
➡️ Date/time add–subtract করতে

```
now + timedelta(days=5)
```

5️⃣ দুইটা date এর difference কীভাবে বের করো?

```
d2 - d1   # returns timedelta
```

6️⃣ Unix Timestamp কী?

A:
➡️ 1970-01-01 থেকে seconds count

```
datetime.now().timestamp()
```

7️⃣ `weekday()` vs `isoweekday()`

A:

- `weekday()` → Monday = 0
- `isoweekday()` → Monday = 1

8️⃣ String থেকে date convert করো কীভাবে?

```
datetime.strptime("2026-01-20", "%Y-%m-%d")
```

9️⃣ Date format change কীভাবে করো?

```
now.strftime("%d-%m-%Y")
```

1️⃣0️⃣ `datetime` object immutable কেন?

A:
➡️ Safety & consistency বজায় রাখার জন্য
➡️ পরিবর্তন করলে নতুন object তৈরি হয়

1️⃣1️⃣ `datetime` comparison করা যায়?

A:
➡️ হ্যাঁ

```
if d1 < d2:
    pass
```

1️⃣2️⃣ Timezone-aware vs naive datetime ⭐⭐⭐

Q: Difference?

A:

- **Naive** → timezone info নাই
- **Aware** → timezone info আছে

```
datetime.now()          # naive
timezone.now()          # aware (Django)
```

1️⃣3️⃣ Django-তে কোনটা use করবে?

A:
➡️ `django.utils.timezone.now()`

1️⃣4️⃣ ISO format কী?

A:
➡️ Standard date-time format (API)

```
now.isoformat()
```

1️⃣5️⃣ `datetime` vs `time` module

A:

- `datetime` → date + time manipulation
- `time` → sleep, timestamp, low-level time

⭐ Top 5 MUST REMEMBER (Interview)

1. `strftime()` vs `strptime()`
2. `timedelta`
3. Unix timestamp
4. Timezone-aware datetime
5. Date comparison

🔥 Bonus Trick (Interview Impression)

```
from datetime import datetime, timezone
datetime.now(timezone.utc)
```